
The Tom Klootwijk Manifold

Bounded-Query Formal Specification and GPU Reference Implementation

Replacement specification v3.0 – PUBLIC edition

Formal verdict. The corrected construction is factually correct under the explicit fixed-profile, finite-precision and accuracy assumptions stated in this document. A single indexed query is worst-case $O(1)$ with respect to represented history/sample count. Literal runtime $o(1)$ is not a correct claim. The valid little-o statement concerns the *normalized encoded-to-dense storage ratio* under fixed or sublinear novelty.

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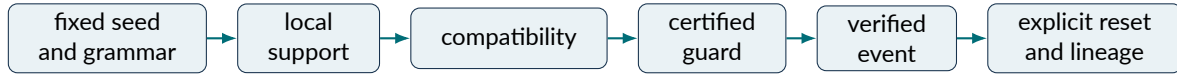
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1 Replacement scope

This specification replaces the earlier Unity, Vulkan-torus and spatiotemporal-SDF interpretations. It is derived from the requester-supplied UGTS-KC 2.0 document and the supplied KLB SeedChain GPU 0.3.0 implementation. It keeps the implementable computer-science core and removes jokes, unsupported physical interpretations, hidden rendering assumptions and incorrect uses of asymptotic notation.

The source architecture is strongest when read as a typed query algebra:



The “seven perpendicular lines” premise is not part of the definition. The source dossier itself identifies it as a comedy-sketch demand. Parity bits and XOR checks remain ordinary finite-data operations; they are not topology-changing physical primitives. An SDF zero is a geometric candidate, not a complete event, transition or identity.

Meaning of the name. “The Tom Klotwijk Manifold” is retained as the requester-selected individual attribution. The name does not by itself establish novelty, priority, ownership, patentability, academic recognition or a new topological homeomorphism class.

2 Mathematical carrier

Let M be a second-countable Hausdorff C^1 manifold of dimension d , equipped with a finite implementation atlas

$$\mathcal{A} = \{(U_a, \varphi_a)\}_{a=1}^A.$$

Let Q be a finite set with the discrete topology. Define

$$X = M \times Q.$$

Since Q is finite and discrete, X is a possibly disconnected C^1 manifold of dimension d : every component $M \times \{q\}$ is a copy of M .

The bundled executable witness uses

$$M = \mathbb{R}^3 \times S^1,$$

so its continuous carrier is four-dimensional. The finite mode label selects a disconnected copy of that carrier. This witness is deliberately modest: it demonstrates the formal contract without claiming that all admissible substrates have this topology.

3 One literal definition

Fix a bounded implementation profile

$$\beta = (J, K, P, H, W),$$

where J bounds model/event segments, K bounds cumulative correction slots per segment, P bounds predictor work or fixed iterations, H bounds guard work, and W fixes machine-word and record widths.

Literal definition.

$$\text{TKM}_\beta(M) := (X, \Sigma, \mathcal{G}, \Phi, \text{Seg}_J, \Delta_K, \text{S, C, R, T, I, L; } \beta), \quad X = M \times Q.$$

The components are:

- Σ : fixed-size seed records;
- $\mathcal{G}$: a finite typed grammar/operator family;

- $\Phi(s, i, t) \in M$: a closed-form or fixed-iteration predictor;
- Seg_J : at most J ordered model/event segments;
- Δ_K : at most K cumulative chart-local corrections per segment;
- S : support predicates;
- C : compatibility predicates;
- R : fields/guards with explicit numerical-error contracts;
- T : explicit reset or transition maps;
- I : topology, chart, continuity and consistency invariants;
- L : lineage and external-novelty metadata.

This is a structure *over* a manifold, not a claim that a binary file, hash chain, GPU buffer or SDF is itself a smooth manifold.

4 State semantics

Let $I_j = [a_j, b_j)$ denote segment j . Admissibility requires the active intervals to be ordered and non-overlapping, so the selector

$$j(t) = \min\{j : t \in I_j\}$$

is unique whenever t lies in the archive domain.

For entity i with seed s_i , the state is

$$x_i(t) = \text{Retr}_{\Phi(s_i, i, t)}(\Delta_{j(t), i}(t)).$$

Here Retr is a declared retraction on M . In the bundled witness,

$$\text{Retr}_{(p, \theta)}(u, \alpha) = (p + u, [\theta + \alpha]_{2\pi}),$$

which is exact for $\mathbb{R}^3 \times S^1$.

The reference predictor is a quadratic jet:

$$\begin{aligned} p(t) &= p_0 + v_0 t + \frac{1}{2} a_0 t^2, \\ \theta(t) &= [\theta_0 + \omega_0 t + \frac{1}{2} \alpha_0 t^2]_{2\pi}. \end{aligned}$$

A segment correction uses the same fixed polynomial degree around its declared origin. This is exact evaluation of the stored model; it is not a claim that quadratic motion is an exact model of arbitrary physics.

4.1 Topology and mode changes

Continuous correction is permitted only inside a declared chart/retraction domain. A mode or topology change is represented by an explicit reset map

$$T_e : G_e \subseteq X \longrightarrow X,$$

not by silently changing the meaning of an SDF, a parity bit or a memory address. If a transition is discontinuous, it is treated as a reset event rather than a continuous zero crossing.

5 Fields, signed distance and event certification

A scalar field $g : X \rightarrow \mathbb{R}$ is not automatically a signed distance function. For a closed domain $\Omega \subset \mathbb{R}^m$, the exact Euclidean signed distance convention used here is

$$d_\Omega(x) = \text{dist}(x, \Omega) - \text{dist}(x, \mathbb{R}^m \setminus \Omega).$$

Then $d_\Omega < 0$ inside, $d_\Omega = 0$ on $\partial\Omega$, and $d_\Omega > 0$ outside. The bundled guard library provides exact sphere SDFs and plane SDFs whose normals are validated to unit length.

Suppose numerical evaluation returns $\hat{g}_k$ with certified absolute error ϵ_k at t_k . A certified endpoint sign separation is

$$\text{Sep}((\hat{g}_0, \epsilon_0), (\hat{g}_1, \epsilon_1)) := [\hat{g}_0 - \epsilon_0 > 0 \wedge \hat{g}_1 + \epsilon_1 < 0] \\ \vee [\hat{g}_0 + \epsilon_0 < 0 \wedge \hat{g}_1 - \epsilon_1 > 0].$$

If $g(x_i(t))$ is certified continuous on $[t_0, t_1]$, the intermediate value theorem then establishes at least one zero in that interval.

A verified transition candidate requires

$$\text{Verified} = \text{IntervalCert} \wedge \text{Support} \wedge \text{Compatible} \wedge (\epsilon \leq \mu) \wedge \text{Sep}.$$

IntervalCert certifies continuity plus support/compatibility validity over the whole interval, not merely at its endpoints. The transition or reset is applied only by the declared policy T.

A tangency, endpoint interval containing zero, or discontinuous reset is therefore not mislabeled as a certified continuous crossing.

6 The constant-query theorem

Let N denote the number of dense samples that could have been materialized, n the total number of entities, and H_{hist} the total history size. A single query is supplied with an indexed entity, time and archive/profile.

For fixed β , the bundled hot path performs:

- one direct seed access;
- exactly/boundedly J segment-slot tests;
- exactly/boundedly K correction-slot tests;
- one fixed-expression predictor and one retraction;
- a bounded guard family.

Therefore

$$T_\beta(N, n, H_{\text{hist}}) \leq c_\Phi + Jc_{\text{seg}} + Kc_\Delta + Hc_g + c_{\text{ret}} =: c_\beta.$$

The right-hand side contains no N , n or H_{hist} , so

$$T_\beta(\text{one indexed query}) = O(1)$$

with respect to those variables.

The constant belongs to the selected profile. Raising J , K , predictor iterations or guard count changes c_β . Cache misses, GPU occupancy, instruction latency, launch overhead and transfer costs affect wall-clock performance without changing this asymptotic statement.

For q independent queries and q output states,

$$T_\beta(q) = \Theta(q), \quad B_{\text{output}}(q) = \Theta(q).$$

No implementation can materialize an unbounded number of outputs in constant total work or zero memory.

7 The legitimate little-o statement

Lowercase little-o describes a quantity that tends to zero relative to a comparison function. It is not a synonym for “very fast” or for ordinary constant time.

Fix a continuous represented interval and a fixed entity count n . Let N be the number of dense samples used only as a comparison. Let

$$B_N = B_0 + b_\nu \nu(N)$$

be encoded bytes, where B_0 is the fixed seed/profile archive and $\nu(N)$ is external novelty that cannot be generated from the model. The dense materialization is

$$D_N = Nnb_x.$$

If

$$\nu(N) = o(N),$$

then

$$\rho_N := \frac{B_N}{D_N} = o(1) \quad (N \rightarrow \infty).$$

For a fixed closed-form archive, $\nu(N) = O(1)$, hence $\rho_N = O(1/N)$.

This is a model-based representation result: increasing the sampling density does not require storing every sample. It is not evidence that an arbitrary pre-existing dense sequence can be losslessly compressed to fixed size. If unpredictable novelty grows as $\Theta(N)$, the ratio need not tend to zero.

Incorrect claim: “nontrivial query runtime is $o(1)$.”

Correct claims: “one indexed query is $O(1)$ under fixed bounds” and “the normalized encoded/dense storage ratio is $o(1)$ under fixed or sublinear novelty.”

8 What the uploaded implementation contributed

The supplied KLB implementation contains several sound engineering ideas:

- fixed-size 64-byte orbit seeds and fixed-trip propagation;
- bounded checkpoint depth in the KLSC1 path;
- sparse novelty records and cumulative checkpoint snapshots;
- host/device-shared functions and one-query-per-thread kernels;
- explicit support, compatibility, guard, event and lineage stages;
- CPU/direct/dense equality checks and documented kill criteria.

Its own documentation correctly warns that model expansion is not lossless compression of a previously materialized trajectory, and that there is no universal $O(1)$, zero-memory or one-cycle guarantee.

8.1 Formal gaps repaired in this package

Issue in KLSC1-style path	Why it blocks the strong claim	Replacement
One 96-byte node per frame	Archive size is $\Theta(N)$ in frame count	Segments occur only at model/event changes
Variable novelty count with binary search	Lookup is $O(\log m)$ unless m is bounded	Fixed K -slot cumulative patch table
Parent traversal in hot query	Work depends on chain depth, even when capped	Cumulative segment snapshot; no parent walk
Implicit overflow behavior	Complexity can change silently	Reject, split or select a new profile
SDF sign used as event shorthand	Zero alone does not certify transition semantics	Interval, error, support and compatibility certificate
Horizon compression ratio	Can be confused with arbitrary lossless compression	Explicit fixed-interval/model-based little-o theorem

9 Reference implementation

The private delivery contains a standalone C++20 project with an optional CUDA executable. No Unity, graphics pipeline, rasterizer, frame clock, ray marcher or TSDF history is required.

9.1 Hot-path data ABI

For the delivered default witness/profile:

Record	Bytes	Role
State	32	position, circle phase, mode, lineage
Seed	64	fixed quadratic predictor coefficients
Patch	64	cumulative segment-local correction
Segment<8>	544	interval, lineage and 8 patch slots
Archive<8,8>	4,368	8 fixed segment records

The exact template profile can be changed at compile time. A profile change is explicit and changes the constant bound.

9.2 Hot query pseudocode

```
state_at(seeds, archive, object, time):
    reject invalid archive or object index
    segment = fixed_scan_J_segments(time)
    base = fixed_predictor(seeds[object], time)
    patch = fixed_scan_K_patch_slots(segment, object)
    state = retract(base, patch)
    return state, segment, patch_hit_count
```

The delivered header uses template constants MaxSegments and MaxPatches. No while loop, parent walk or variable-size binary search occurs in the hot query.

9.3 CUDA mapping

The optional CUDA file launches one thread per query. Each thread executes the same bounded data-flow contract and writes one result. The GPU batch is therefore parallel $\Theta(q)$ work, not total $O(1)$ work. The source compiles only when a supported CUDA toolkit is available; architecture selection is an explicit CMake parameter rather than a hidden hard-coded target.

The runtime code, tests and examples contain no requester identifier or birth date. Those values appear only in the private attribution record.

10 Admissibility and kill criteria

A package claiming the fixed-profile theorem must enforce all of the following:

1. $\text{segment_count} \leq J$ and active patch count per segment $\leq K$.
2. Segment intervals are finite, ordered and non-overlapping, unless an explicit deterministic priority rule is part of the schema.
3. Each entity key occurs at most once per segment patch table.
4. Predictor, patch and guard values are finite and stay inside the declared chart/retraction domain.
5. Plane SDF normals are unit length; sphere radii are positive; approximate fields carry explicit error bounds.
6. Continuous crossings have an interval continuity/support/compatibility certificate.
7. Predictor and field error are below the application event margin.
8. External topology changes use explicit reset/gluing maps.
9. Archive/segment selection that occurs outside the hot query is not hidden inside the complexity claim.
10. If the profile is exceeded, the builder rejects, splits the archive, chooses a larger named profile or uses a variable-cost/dense path.

The $o(1)$ storage-ratio claim must be withdrawn when external novelty becomes $\Theta(N)$, when every sample is stored, or when comparison samples are not generated by the same declared model/error contract.

11 Validation record

11.1 Requester-supplied source package

The source ZIP was unpacked and its internal SHA-256 manifest verified. A clean CPU-only build completed under GCC 14.2.0 and CMake 3.31.6. Both included CTest suites passed. This validates internal C++ behavior and integrity checks, not every physical or orbital claim in the source material.

11.2 Replacement implementation

Check	Result
C++20 release configure/build	PASS
Release CTest	1/1 PASS
Randomized independent state comparisons	20,000 PASS
Exact sphere and normalized-plane SDF tests	PASS
Interval-certificate requirement	PASS
Archive overlap and duplicate-patch rejection	PASS
AddressSanitizer/UndefinedBehaviorSanitizer CTest	1/1 PASS
Fixed-loop structural audit	PASS
Sensitive-token scan of runtime/test/example source	PASS
Clang CUDA host-side syntax parse with declaration stub	PASS
Native nvcc compilation	NOT EXECUTED
Physical NVIDIA GPU execution	NOT EXECUTED

The CUDA result is therefore source-level and syntax-level validation, not a claim of target-GPU execution or performance. No benchmark speedup is asserted.

12 Factual-correctness declaration

Yes - conditionally and precisely. The replacement definition and CPU implementation are factually correct under the fixed profile, finite-precision, continuity, support/compatibility and error-margin assumptions stated here. The bundled CUDA path is a plausible implementation of the same contract, but it was not natively compiled or executed in this preparation environment.

Statement	Verdict	Reason
One indexed query is $O(1)$ under fixed J, K, P, H, W	Correct	All hot-loop and record bounds are profile constants
Total work for q outputs is $\Theta(q)$	Correct	Every output requires at least one query/write
Runtime is literally $o(1)$	Incorrect	Nontrivial discrete work does not tend to zero
Encoded/dense ratio is $o(1)$ for fixed/sublinear novelty	Correct	Direct consequence of $B_N/(Nnb_x) \rightarrow 0$
Arbitrary external history is exact in fixed storage	Incorrect	Unpredictable novelty must be stored or rejected
A field zero alone changes topology or identity	Incorrect	Reset/transition semantics are separate
The label proves a new topological species or authorship	Incorrect	Naming and technical proof are distinct

13 Attribution, privacy and provenance boundary

This public edition masks the requester-supplied identifier and birth date. The private edition and private ZIP contain the full values and should not be published. Runtime headers, CUDA source, tests, examples and ABI records contain no direct identifier.

The private attribution fingerprint is intentionally omitted from this public-sharing edition. Delivery manifests use SHA-256 for file-integrity checking; runtime lineage values remain compact routing/check values rather than durable legal identity.

14 Source register

1. Requester-supplied *UGTS-KC 2.0 – The Unified Geometric-Topological Substrate*, used for the query-first tuple, support/compatibility/guard/event chain, reconstruction boundary and kill criteria.
2. Requester-supplied `klb_seedchain_gpu_v0.3.0(4).zip`, used for fixed-size seed records, host/device code patterns, bounded predictor ideas, sparse novelty, validation discipline and failure-boundary evidence.
3. Earlier requester-supplied Tom Klootwijk dossiers, used only to preserve the requested individual nomenclature and to remove the explicitly comedic perpendicular-lines condition and unsupported parity/SDF claims.

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